Institute of Technology of Cambodia

Department of Applied Mathematics and Statistics

Spam Email Probability Analysis

Presented By Group 3

Subject: Probability
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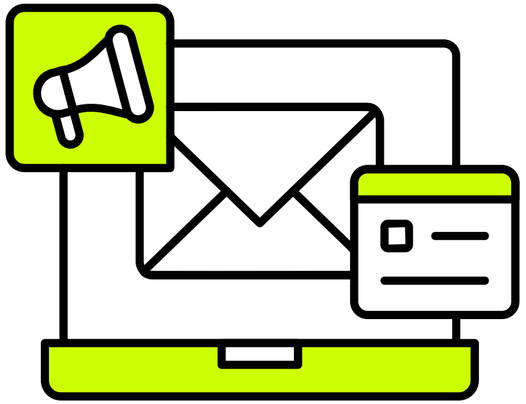
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I.INTRODUCTION

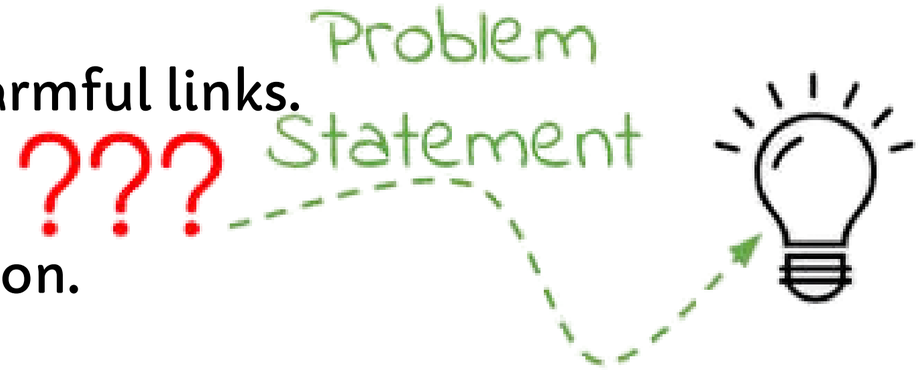


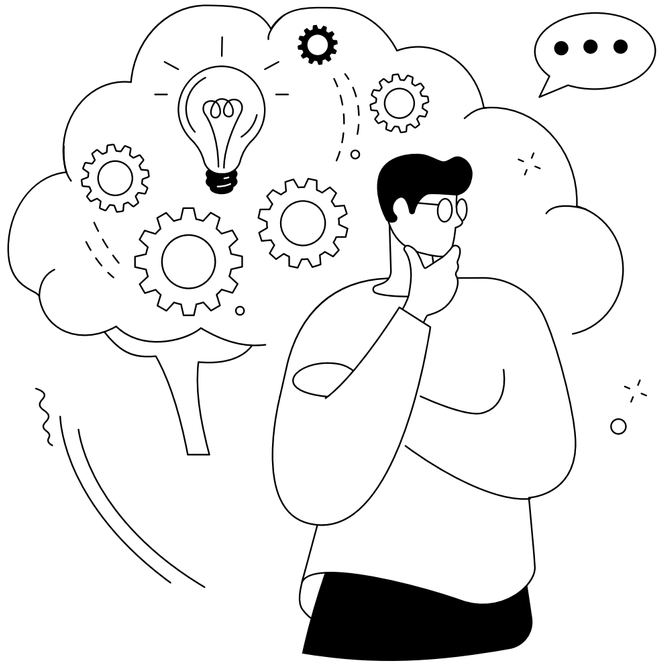
- Spam emails are unwanted messages that are sent to many users through email systems.



1.PROBLEM STATEMENT

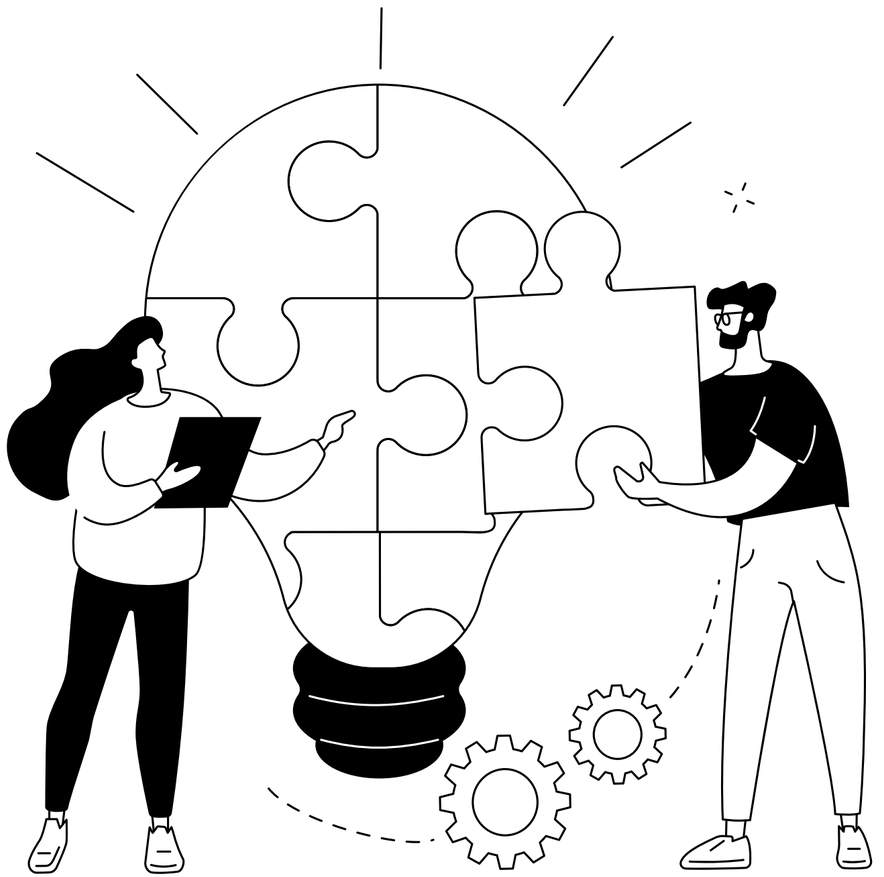
- They may contain scams, fake promotions, or harmful links.
- Spam emails waste time and fill inboxes.
- Some spam emails may steal personal information.



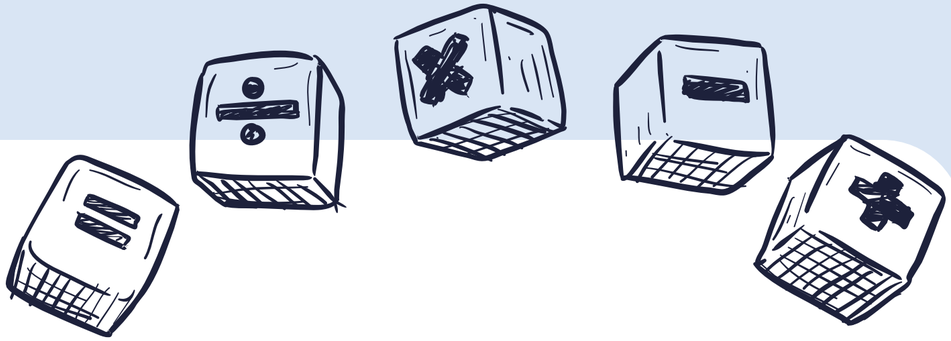


2.IMPORTANCE OF THE PROBLEM

- Email is widely used in schools and businesses.
- Spam emails can create security risks.
- They make communication difficult and unsafe.
- Studying spam helps improve cybersecurity and spam filters.



3.ROLE OF PROBABILITY



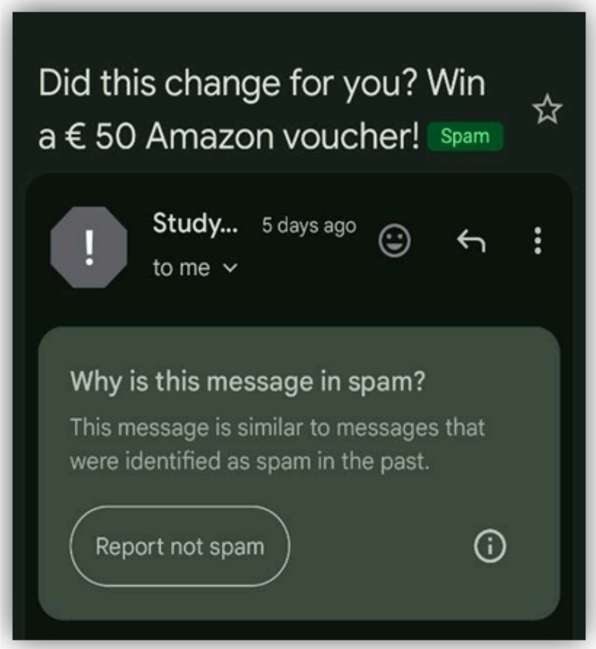
- Probability helps measure spam email occurrences.
- It helps predict spam patterns and behavior.
- This study uses:
  - > Probability Distribution
  - > Expected Value
  - > Variance
  - >Standard Deviation

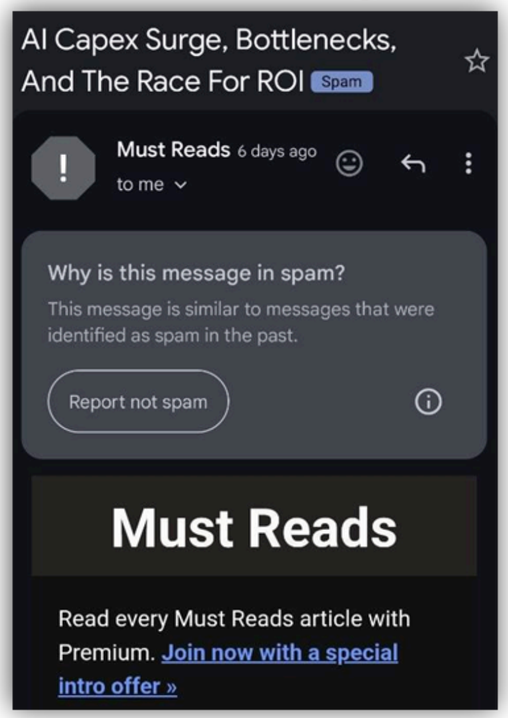


II.DESCRIPTION OF THE EXPERIMENT

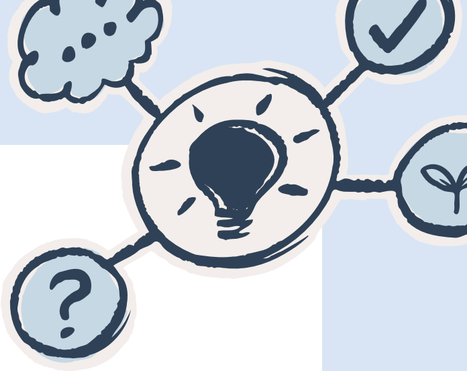
2.1 METHOD OF DATA COLLECTION

- Survey conducted with 30 ITC students.
- Students checked Gmail spam emails.
- Screenshots were collected as evidence.
- 9 students were affected by spam emails.
- Spam categories included:
  - > Education
  - > Ai/Github
  - > Lottery
  - > Game
  - > Storage Image





2.2 DEFINE THE RANDOM VARIABLE



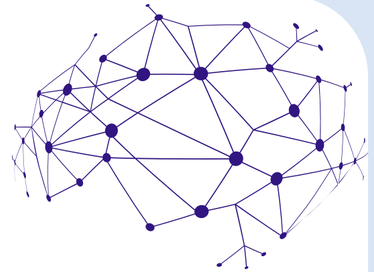
From the collected data, the random variable is defined as:

X = the number of spam categories received by an affected student during the two-weeks survey.

The survey was conducted among 30 students and 9 students were found to have received spam emails.





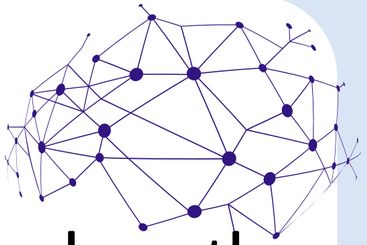


So, the possible values of the random variable are:
$X = \{1, 2, 3, 4\}$
Where:

- X = 1 means a student receives spam from one category only
- X = 2 means a student received spam from two categories
- X = 3 means a student received spam from three categories
- X = 4 means a student received spam from categories

This random variable is a discrete random variable because the number of spam categories can only be counted using whole numbers.

2.3 PROBABILITY DISTRIBUTION



Base on data receives: we have 30 students and we found 9 students who have the different and the same spam emails.

Number of Student	Spam Category
Student 1	Lottery Money, Game, Storage Image
Student 2	Education
Student 3	AI / GitHub, Lottery Money, Education
Student 4	Education
Student 5	Education
Student 6	AI / GitHub
Student 7	Lottery Money, Storage Image
Student 8	Lottery Money
Student 9	AI / GitHub

Probability Distribution



Probability Distribution Table

Number of Spam Categories X	Frequency	Probability P(X)
1	6	$\frac{6}{9} = 0.667$
2	1	$\frac{1}{9} = 0.111$
3	2	$\frac{2}{9} = 0.222$
4	0	$\frac{0}{9} = 0.000$

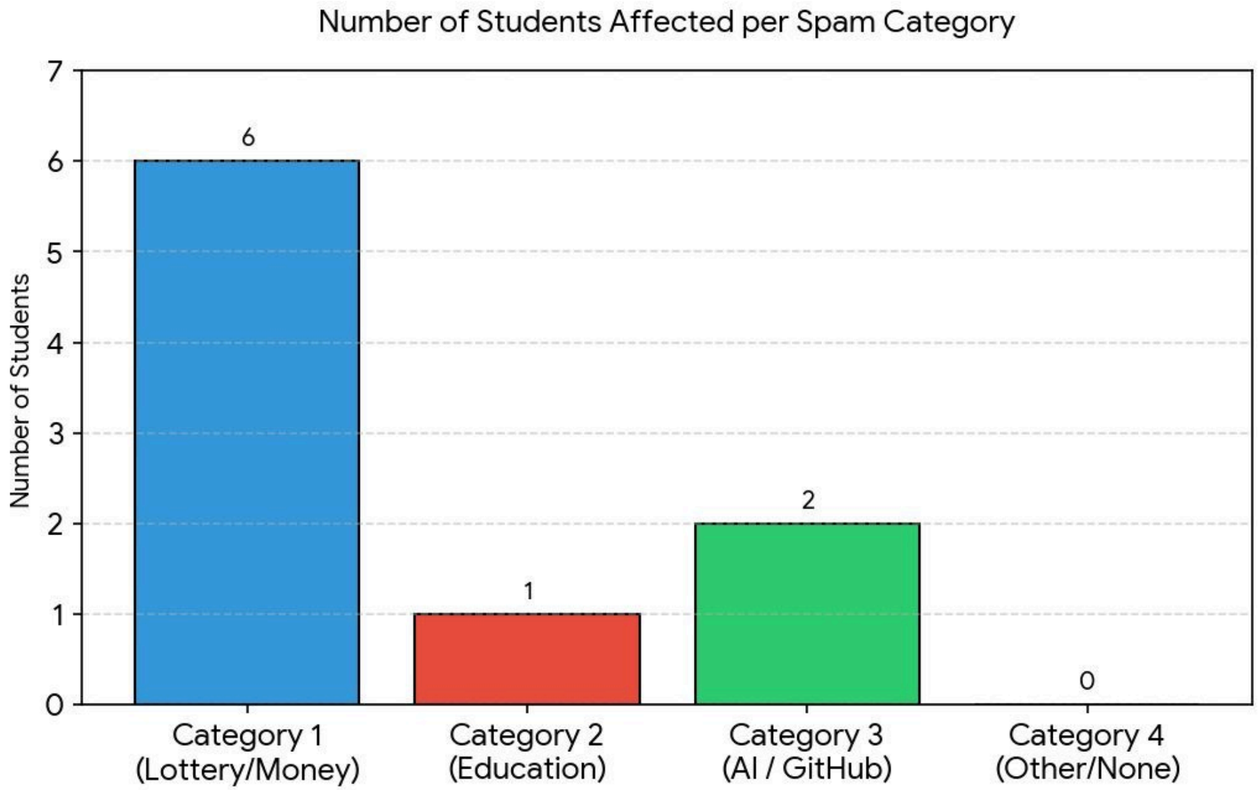
Check:

0.667 + 0.111 + 0.222 + 0.000 = 1.000

The total probability equals 1.000 which confirms that the probability distribution is valid.

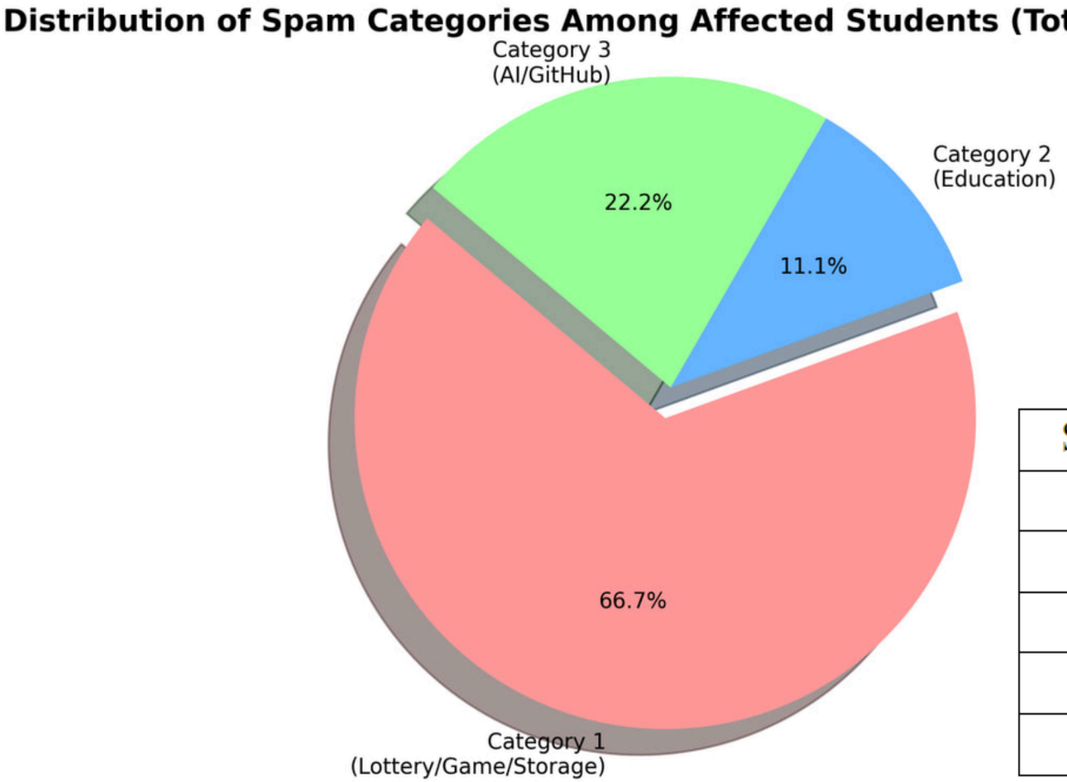
2.4 PROBABILITY CALCULATIONS

Base on data we identity the number of students affect by category spam:



This chart highlights the exact count of students affected by each spam type.

FINDING



$$P(X = x) = \frac{\text{Number of Students affect by spam}}{\text{Total students affects by spam}}$$

Spam category (x)	Number of Students affect by spam	Probability
1	6	6/9 = 0.6667
2	1	1/9 = 0.1111
3	2	2/9 = 0.2222
4	0	0.0000
Total	9 students	1.0000

FINDING

We Let random variable X = be number of spam categories received by the students.
Possible value x can be 1,2,3,4
The events that we would like to study is:

The probability that student(from 30 students) is affected by spam email (9 students)

$$P(\text{affected}) = \frac{\text{number of students affect by spam email}}{\text{total student}}$$

$$P(\text{affected}) = \frac{9}{30} = 0.30 = 30\%$$

This show that a 30% of students are receives the same and also different spam emails.

FINDING

The probability that students is affected has exactly 1 spam email category.

DEFINITION

For any two events A and B with $P(B) > 0$, the **conditional probability of A given that B has occurred** is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \qquad (2.3)$$

$$P(X = 1 \mid \text{affected}) = \frac{\text{number of students affect by spam email} \cap \text{student got 1 category}}{\text{total student affects by spam}}$$

$$P(X = 1 \mid \text{affected}) = \frac{6}{9} = 0.6667 = 66.67\%$$

This show that a 66.67% of students are receives the only one spam email category.

FINDING

The probability that the student has 2 or 3 spam emails categories.

$$P(2 \leq X \leq 3 \mid \text{affected}) = \frac{\text{number of students affect by spam email} \cap \text{student got 2 or 3 category}}{\text{total student affects by spam}}$$

$$P(X = 2 \cup X = 3 \mid \text{affected}) = \frac{1}{9} + \frac{2}{9} = 0.1111 + 0.2222 = 0.3333 = 33.33\%$$

This show that a 33.33% of students are receives 2 or 3 spam emails categories.

FINDING

The probability that students receive Education spam.

Base on data we have 4 students who receive Education spam.

$$P(\text{Education} \mid \text{affected}) = \frac{\text{number of students affect by spam email} \cap \text{student receive Education spam}}{\text{total student affects by spam}}$$

$$P(\text{Education} \mid \text{affected}) = \frac{4}{9} = 0.4444 = 44.44\%$$

This show that a of students are receives Education spam category.

2.5 EXPECTED VALUE

We calculate the expected number of spam categories of affected student.
X is number of students who receive spam emails.

The Expected Value of *X*

DEFINITION

Let X be a discrete rv with set of possible values D and pmf $p(x)$. The **expected value** or **mean value** of X, denoted by $E(X)$ or μ_X or just μ, is

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

Expected value form by:

$$E(X) = \sum x \cdot P(X = x)$$

$$E(X) = 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4)$$

$$E(X) = 1 \cdot \left(\frac{6}{9}\right) + 2 \cdot \left(\frac{1}{9}\right) + 3 \cdot \left(\frac{2}{9}\right) + 4 \cdot \left(\frac{0}{9}\right) = \frac{14}{9} \approx 1.5556$$

2.6 VARIANCE AND STANDARD DEVIATION

DEFINITION

Let X have pmf $p(x)$ and expected value μ. Then the **variance** of X, denoted by $V(X)$ or σ_X^2, or just σ^2, is

$$V(X) = \sum_D (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The **standard deviation** (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

A Shortcut Formula for σ^2

The number of arithmetic operations necessary to compute σ^2 can be reduced by using an alternative formula.

PROPOSITION

$$V(X) = \sigma^2 = \left[\sum_D x^2 \cdot p(x)\right] - \mu^2 = E(X^2) - [E(X)]^2$$

VARIANCE AND STANDARD DEVIATION

- Variance: $V(X) = E(X^2) - [E(X)]^2$

Find $E(X^2)$

$$E(X^2) = \sum x^2 \cdot P(X = x)$$

$$E(X^2) = 1^2 \cdot P(X = 1) + 2^2 \cdot P(X = 2) + 3^2 \cdot P(X = 3) + 4^2 \cdot P(X = 4)$$

$$E(X^2) = 1 \cdot \left(\frac{6}{9}\right) + 4 \cdot \left(\frac{1}{9}\right) + 9 \cdot \left(\frac{2}{9}\right) + 4 \cdot \left(\frac{0}{9}\right) = \frac{28}{9} \approx 3.1111$$

$$V(X) = 3.1111 - (1.5556)^2 \approx 0.6912$$

- Standard Deviation: $V(X) = \sigma_X^2$

Then $\sigma_X = \sqrt{V(X)} = \sqrt{0.6912} \approx 0.8313$

III.CONCLUSIONS

30%
of students received spam emails in two weeks

E(X)=1.56
average spam categories per student

66.7%
of affected students got spam from only 1 category

σ=0.831
Low variability consistent behavior

MAIN FINDINGS

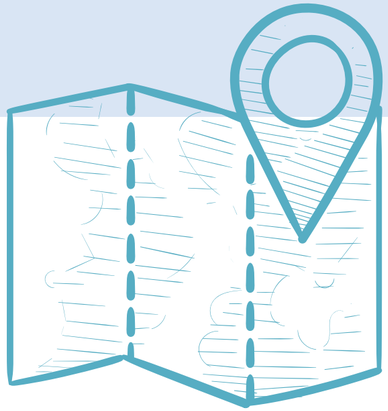
- Spam emails are a common issue, affecting nearly 30% of ITC students in just two weeks.
- Most affected students received spam from only one category the most frequent outcome (P(X=1) = 0.667).
- Low variance (V(X) = 0.691) confirms that the number of spam categories is stable and predictable.
- Education and AI/GitHub were the most common spam categories observed in the survey.

OVERALL TAKEAWAYS

- Spam emails waste time, fill inbox storage, and carry phishing and scam risks for students.
- The clear pattern in spam behavior can guide the development of better spam filters and user protections.
- This project successfully applied probability distribution, conditional probability, expected value, variance, and standard deviation to a real-world problem.



IV.REFERENCES



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THANK YOU!

Q&A!